Sharif University of Technology

Physics-Informed Modeling of Fluid Flow and Particle Transport in Curved Channels

Research Proposal

Master of Science

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Preliminary computational evidence and proposed research plan

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Abstract

Reliable prediction of particle transport in curved channels requires both an accurate fluid field and a faithful treatment of particle inertia and wall contact. This proposal develops a physics-informed neural-network workflow for steady incompressible flow and dilute spherical-particle transport, with independent numerical references and explicit accuracy criteria. A flow network with five hidden layers of 70 neurons predicts velocity and pressure; a force network with five hidden layers of 90 neurons approximates the specified Stokes-drag relation. Existing repository results provide preliminary evidence: the principal curved-channel case has velocity and pressure errors of 1.510% and 0.863% relative to an independently assembled finite-element reference. Across two training seeds, the mean velocity error is 1.469%. However, only 6 of 7 baseline particle endpoint classes agree with the reference, and some trajectory errors exceed the current screening limits. A separate three-dimensional manufactured benchmark gives velocity and pressure errors of 2.428% and 13.501%, respectively. These findings motivate a research program focused on near-wall accuracy, trajectory-based evaluation, controlled geometry studies, and pressure recovery. The proposed study will separate numerical verification from physical validation and will evaluate generalization only on explicitly held-out cases. The expected contribution is a reproducible, bounded assessment of when physics-informed flow approximations support reliable particle-transport predictions.

Keywords: physics-informed neural networks; particle transport; curved channels; Stokes drag; numerical verification.

Document and Evidence Record

This is a research proposal supported by existing preliminary computations. Completed numerical results appear in chapter 6; the experiments and improvements in chapter 7 are proposed work. No experimental measurements or completed three-dimensional channel study are claimed.

The document adapts the layout, typography, margins, front matter, and chapter-based structure of the supplied English thesis template. The original template is retained with the source. Thesis approval, originality-signature, and completion pages are omitted because they do not describe this proposal. The author and supervisor details were supplied for the title page.

Numerical tables and highlighted values are generated from `results/final_study/metrics.json` and associated CSV/JSON files. All 21 research figures are read from the repository's existing PDF exports. The evidence data revision is `97d71480fc01`. A machine-readable input hash record accompanies the source. The document build does not retrain models or alter numerical results.

Repository:

`https://github.com/mohrezsayeh90-ai/basic_particle_problem`

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Abbreviations and Nomenclature

Abbreviation	Meaning
ABC	Arnold–Beltrami–Childress manufactured flow
CFD	Computational fluid dynamics
FEM	Finite-element method
MLP	Multilayer perceptron
ODE / PDE	Ordinary / partial differential equation
PINN	Physics-informed neural network
RMS	Root-mean-square

Symbol	Meaning
L_0, U_0, T_0	Reference length, speed, and time; $T_0 = L_0/U_0$
$\boldsymbol{u}, p$	Dimensionless fluid velocity and pressure
$\boldsymbol{x}_p, \boldsymbol{v}_p$	Dimensionless particle position and velocity
$R_{\text{in}}, R_{\text{out}}$	Nominal inner and outer radii of the arc channel
h, W, L	Channel half-width, width $W = 2h$, and centerline length
s, η	Centerline coordinate and normalized transverse coordinate
d_p, ρ_r	Dimensionless particle diameter and density ratio ρ_p/ρ_f
$\mathrm{Re}, \mathrm{Re}_p, \mathrm{St}$	Reference Reynolds number, particle Reynolds number, Stokes number
$\boldsymbol{F}^*$	Drag force scaled by displaced-fluid mass times U_0^2/L_0
θ, ϕ	Parameters of the flow and force networks
Q, Q_0	Section flux and prescribed reference flux

Chapter 1

Introduction

1.1 Background and Problem

Curved channels couple fluid turning with particle inertia. A fluid marker follows the local velocity, whereas a finite-inertia particle can lag behind a change in flow direction and approach a wall. Predicting the resulting trajectory is relevant to transport, separation, and deposition studies. The mathematical difficulty is broader than approximating a smooth velocity field: small, spatially localized errors can change a discrete outcome such as whether a particle exits or sticks.

Physics-informed neural networks incorporate governing-equation and boundary residuals into a learning objective [1, 2]. They offer a differentiable approximation that can be evaluated during trajectory integration. Their optimization, scaling, and accuracy nevertheless require verification; a small loss or an attractive plot is insufficient evidence of a reliable solution [3, 4]. Classical CFD remains a necessary independent comparator in the present study.

The current project already includes a two-network implementation, analytical tests, an independent FEM reference, and particle tracking. Its preliminary results reveal a useful research problem: the principal flow approximation has a relatively small mean error, yet some particle paths and endpoint classes differ from the reference. This proposal therefore studies the connection between field accuracy and transport accuracy, rather than assuming one implies the other.

1.2 Aim and Objectives

The aim is to develop and assess a reproducible physics-informed workflow for predicting dilute-particle transport in smooth curved channels within an explicitly defined physical and geometric scope.

O1. Quantify flow and pressure accuracy against analytical and independently assembled numerical references, including mesh and interpolation sensitivity.

O2. Determine how local field errors, inertia, particle size, and wall clearance affect trajectory errors and endpoint classification.

O3. Evaluate targeted changes to training and sampling using separate development and final evaluation cases.

O4. Establish the supported scope of geometry variation and a distinct three-dimensional verification pathway, without inferring universal generalization from the network input dimensions.

1.3 Research Questions

RQ1. Under which training and reference-resolution conditions does the flow PINN satisfy both field-error and mass-conservation criteria?

RQ2. How does flow-approximation error propagate into particle position, residence time, and sticking or exit decisions?

RQ3. Does a learned force surrogate materially change trajectories compared with exact Stokes drag evaluated on the same learned flow?

RQ4. Which conclusions persist across separately trained channel shapes and Reynolds numbers, and what further evidence is needed for a 3D channel model?

The working expectation is that near-wall and turning-region accuracy will be more informative for deposition than a global sampled velocity norm. This is a testable hypothesis for the proposed study, not a mechanism already established by the preliminary figures.

1.4 Scope and Intended Contribution

The initial physical model is steady, incompressible, Newtonian flow with one-way coupling to dilute, heavy, spherical particles. Stokes drag and irreversible wall sticking are retained. The supported channel family has a smooth, nonintersecting planar centerline and constant width. Sharp corners, junctions, variable-width networks, particle–particle forces, and feedback on the fluid are outside the present implementation.

The intended contribution is a reproducible evaluation linking field errors to transport errors. Its scope excludes a universal drag law and arbitrary 3D geometry. The remaining chapters develop the foundations, methods, evaluation, preliminary evidence and proposed work.

Chapter 2

Literature and Research Rationale

2.1 Physics-Informed Flow Approximation

Raissi et al. formulate PINNs by penalizing differential-equation residuals alongside observed or prescribed information [1]. The broader physics-informed learning framework includes several ways to combine physical constraints and learned approximations [2]. For this project, the relevant distinction is between a network trained on PDE and boundary information and a regression model trained directly on interior CFD labels. The current flow model uses the former design, and its FEM interior samples are reserved for numerical comparison.

Optimization behavior limits this approach. Gradient imbalance among loss terms and unequal learning rates of solution components have been studied in PINNs [3, 4]. These results motivate scaling the momentum and continuity residuals, retaining fixed monitoring samples, and reporting boundary and flux errors separately. They do not establish that any particular optimizer or architecture will be accurate in the proposed curved-channel problem.

2.2 Independent Flow and Particle References

Mixed finite elements provide a classical route to incompressible flow. The Taylor–Hood construction uses a higher-order velocity space than pressure space [5]; the present independent reference uses P2 velocity and P1 pressure with scikit-fem [6]. Independence here means that this solver is assembled without PINN predictions. Its own nonlinear, mesh, and continuity checks remain necessary. It is a project-specific CFD reference, not a claim that commercial CFD or an experimental benchmark was used.

The full small-sphere equation of motion includes terms beyond instantaneous Stokes drag [7]. The present model deliberately uses a heavy-particle simplifica-

tion. This is appropriate only within a qualified regime and cannot represent every particle–fluid interaction. In particular, geometric wall contact does not supply the hydrodynamic wall corrections that may become important at small clearances.

2.3 Verification, Validation, and the Specific Gap

Numerical verification addresses consistency with the stated equations; physical validation concerns agreement with suitable external physical evidence [8]. Analytical Poiseuille flow and a manufactured three-dimensional solution are strong implementation checks. Agreement with a refined FEM solution tests numerical consistency for the same model. Neither substitutes for experiment when making device-level predictions.

Table 2.1: Roles of the main methodological foundations.

Foundation	Role in this proposal	Limit of inference
PINNs [1, 2]	Differentiable flow approximation constrained by physics	Training loss does not certify trajectory accuracy.
Training analysis [3, 4]	Motivation for loss scaling and separate diagnostics	Remedies must be evaluated on this channel family.
Taylor–Hood FEM [5]	Independent velocity/pressure reference	Discretization and geometry errors remain.
Small-sphere dynamics [7]	Basis for the drag-only heavy-particle model	Other force terms and confinement are omitted.
Numerical verification [8]	Reference hierarchy and error reporting	Numerical agreement is not experimental validation.

The proposed gap is local to the current workflow: its global field metrics and particle-outcome metrics do not yet support the same level of confidence. The study will investigate that discrepancy and determine a defensible operating envelope. This is a focused narrative rationale based on the cited foundational work and repository evidence; it is not presented as an exhaustive or systematic survey of all particle-transport PINNs.

Chapter 3

Physical and Mathematical Framework

3.1 Geometry and Nondimensionalization

The default S-channel joins two semicircular segments with a continuous tangent and piecewise curvature. Its nominal radii are $R_{\text{in}} = 0.20$ and $R_{\text{out}} = 0.50$; the centerline radius is their average. The width is $W = R_{\text{out}} - R_{\text{in}} = 0.30$ and the centerline length is approximately $L = 2.1991$. A general supported channel is represented by

$$\boldsymbol{x}(s,\eta) = \boldsymbol{c}(s) + h\eta\boldsymbol{n}(s), \qquad 0 \le s \le L, \quad -1 \le \eta \le 1, \quad h = W/2, \tag{3.1}$$

where $\boldsymbol{c}$ is a planar centerline and $\boldsymbol{n}$ its unit normal. Curvature and overlap checks exclude invalid offset tubes. A custom CSV changes the centerline, not the physical topology supported by the solver.

Dimensional variables are scaled using L_0, U_0 and $\mu U_0/L_0$ for length, speed and pressure. Time is scaled by $T_0 = L_0/U_0$. Throughout the following equations, variables without a dimensional subscript are dimensionless. The reference Reynolds number is $\mathrm{Re} = \rho_f U_0 L_0/\mu$; L_0 is not implicitly a hydraulic diameter. The current principal case uses $\mathrm{Re} = 1.0$.

3.2 Fluid Equations and Boundary Conditions

For a steady incompressible Newtonian fluid,

$$\nabla \cdot \boldsymbol{u} = 0, \tag{3.2}$$

$$\mathrm{Re}(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \nabla^2 \boldsymbol{u}. \tag{3.3}$$

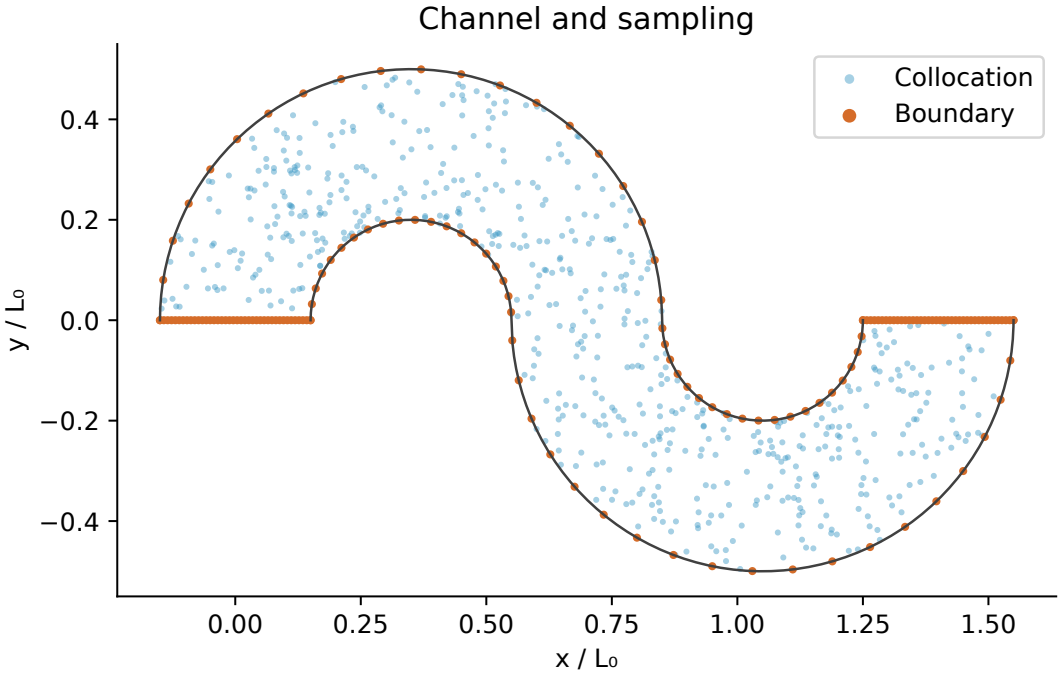


Figure 3.1: Default channel and sampled interior and boundary points from the repository. Coordinates are nondimensional; the displayed point set illustrates the sampling procedure.

The walls satisfy no slip. Both end sections have the prescribed tangential velocity $\boldsymbol{u} = (1 - \eta^2)\boldsymbol{t}$ with unit center speed. Pressure has one gauge condition. In particular, the outlet is a prescribed-velocity boundary, rather than an assumed zero-pressure outlet. The FEM and PINN use these same physical boundary conditions, with their pressure constants aligned during comparison. The prescribed section flux is $Q_0 = 4h/3$.

3.3 Particle and Fluid-Marker Dynamics

Let d_p be particle diameter and $\rho_r = \rho_p/\rho_f$ the density ratio. The adopted drag-only heavy-particle equations are

$$\frac{d\boldsymbol{x}_p}{dt} = \boldsymbol{v}_p, \tag{3.4}$$

$$\frac{d\boldsymbol{v}_p}{dt} = \frac{18}{\rho_r \mathrm{Re} d_p^2}[\boldsymbol{u}(\boldsymbol{x}_p) - \boldsymbol{v}_p] = \frac{\boldsymbol{u}(\boldsymbol{x}_p) - \boldsymbol{v}_p}{\mathrm{St}}, \qquad \mathrm{St} = \frac{\rho_r \mathrm{Re} d_p^2}{18}. \tag{3.5}$$

The particle Reynolds number is $\mathrm{Re}_p = \mathrm{Re} d_p \|\boldsymbol{u} - \boldsymbol{v}_p\|$. Stokes drag requires $\mathrm{Re}_p \ll 1$ and a particle small relative to flow length scales. The force formulation is a simplification of small-sphere dynamics, not the full Maxey–Riley equation [7].

A fluid marker follows $d\boldsymbol{x}/dt = \boldsymbol{u}(\boldsymbol{x})$. A finite-size particle sticks when its center reaches a distance $d_p/2$ from the wall; its velocity is then set to zero. This contact rule

is irreversible and does not represent rebound or adhesion-force calibration. Initial particle velocity equals the local fluid velocity unless the user supplies a different admissible value.

3.4 Force-Network Interpretation

The normalized drag force $\boldsymbol{F}^*$ is scaled by displaced-fluid mass times U_0^2/L_0. With input $z = (u, v, p, d_p, x_p, y_p, u_p, v_p)$ and normalized input map N, the implemented surrogate is

$$\widehat{\boldsymbol{F}}^*(z) = \frac{18}{\mathrm{Re} d_p^2}\{g_\phi(Nz) - g_\phi(Nz_0)\}, \tag{3.6}$$

where z_0 replaces the particle velocity by the fluid velocity. Subtraction enforces zero force at zero slip. The network is trained against the algebraic Stokes residual, so it approximates an already specified law. Pressure and particle position are input slots; they do not establish additional force mechanisms. Exact Stokes drag remains the reference.

3.5 Physical Limits

The model omits gravity, lift, Brownian motion, added mass, history effects, fluid feedback, particle collisions, and near-wall lubrication corrections. The largest tested diameter is 15% of the channel width, so that case is a sensitivity study rather than proof of the unbounded point-particle model's physical accuracy. The computed maximum Re_p is 0.01221; this supports a low-slip-inertia regime but does not remove confinement or dilute-loading assumptions.

Chapter 4

Computational Methodology

4.1 Architecture and Training

The flow network maps $(x, y, R_{\text{in}}, R_{\text{out}}, \text{Re})$ to (u, v, p) through five hidden layers of 70 tanh units. The force network maps the eight inputs in equation (3.6) to two force components through five hidden layers of 90 tanh units. Spatial coordinates are affinely normalized, and the pressure output uses a scale $p_s = 2L/h^2$.

For continuity residual r_c and the two momentum residuals $\boldsymbol{r}_m$, the implemented flow objective is

$$\begin{aligned}\mathcal{L}_{\text{flow}} = \; & \text{MSE}(hr_c) + \text{MSE}(h^2\boldsymbol{r}_m) + 20\,\text{MSE}(\widehat{\boldsymbol{u}} - \boldsymbol{u}_{\text{BC}}) \\ & + (\widehat{p}_g/p_s)^2 + 5\,\text{MSE}(Q/Q_0 - 1).\end{aligned} \tag{4.1}$$

Here the gauge pressure target is zero, and flux is sampled on seven sections using 25 transverse points per section. This integral mass constraint helps exclude a low-throughput interior field that would be inconsistent with the prescribed inlet flux. The force loss is the mean-square residual between normalized predicted drag and fluid–particle slip.

Interior samples are resampled during Adam training. Fixed PDE-monitor samples support checkpoint selection, followed by L-BFGS refinement on fixed collocation samples. Reference FEM interior labels do not enter the flow-training objective. Each geometry/Reynolds-number case is trained separately; fixed parameter input slots are not evidence of a parametric solution across untrained geometries.

These settings describe the original training recorded in the checkpoint metadata. The archived final study reloaded those weights to reproduce evaluation and figures; its default command-line preset fields should not be mistaken for a second training run.

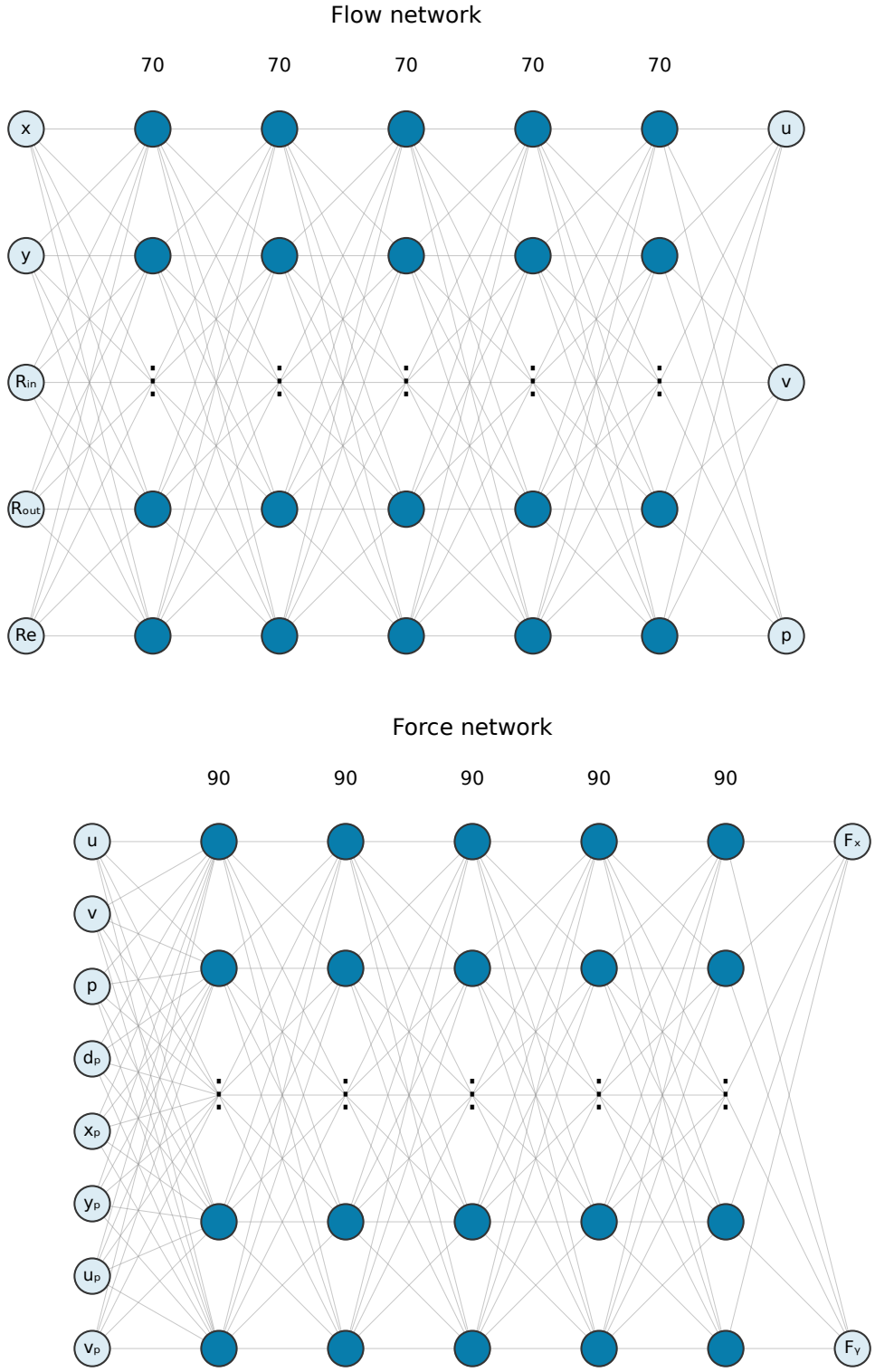


Figure 4.1: Flow network (top) and force network (bottom), shown as enlarged crops of the existing architecture PDF. Five hidden layers contain 70 and 90 units, respectively. Dots abbreviate neurons; labels give actual layer widths.

Table 4.1: Recorded settings of the preliminary training run.

Item	Recorded setting
Principal flow seeds	42 and 43; first seed used for the principal comparison
Initial flow phase	1,000 Adam epochs, 512 interior samples per epoch, 250 L-BFGS iteration budget
Refinement phase	1,800 Adam epochs, 1,024 samples per epoch, 700 L-BFGS iteration budget
Adam learning rate	10^{-3} with cosine scheduling to 5×10^{-5} per flow phase
Force training	1,000 Adam epochs; batches of 512; learning rate 8×10^{-4}
Force selection and testing	1,024 selection samples; separately sampled held-out tests
Flow evaluation	1,600 fixed interior points for direct FEM comparison
3D manufactured benchmark	800 Adam epochs and 400 L-BFGS iteration budget; a separate model

4.2 Independent FEM Reference

The reference solves equations (3.2) and (3.3) on a triangular mesh constructed from the same centerline and width. Vector P2 velocity and P1 pressure spaces form the Taylor–Hood discretization [5, 6]. Picard iteration treats the nonlinear advection term, and each mixed linear system is solved with a sparse direct solver. The finest recorded mesh has 3026 triangles. Identical velocity conditions are imposed at the walls and both end sections; the arbitrary pressure constant is fixed separately.

The curved wall is polygonal in the reference mesh, so mesh refinement changes both discretization and geometric approximation. Three resolutions are retained rather than treating a single successful solve as exact. For repeated trajectory evaluations, a fast interpolator operates in the channel's longitudinal/transverse coordinates. This prevents interpolation across the hole inside a bend. Its error is measured against direct FEM evaluation before trajectory comparisons are interpreted.

4.3 Trajectory Integration and Cases

ODE integration uses RK45 when the relaxation time is sufficiently long relative to the geometric step limit, and Radau for shorter, stiffer relaxation scales. The current tolerances are 2×10^{-6} relative and 2×10^{-8} absolute. Event functions detect wall contact, outlet crossing and inlet backflow; time and evaluation budgets keep each run bounded. A failed path remains marked as failed. A finite output is never silently replaced by the reference trajectory.

The principal particle has $d_p = 0.025$ and $\rho_r = 1000$. Seven inlet positions span $\eta = -0.65$ to 0.65. Diameter cases use $d_p = 0.0105, 0.0255, 0.045$, and density cases use

$\rho_r = 200, 1000, 3000$. The baseline and the $\rho_r = 1000$ control are deliberately repeated in the output records and are not independent physical replications.

Three comparisons isolate different error sources:

1. PINN flow with ForceNet versus FEM flow with exact Stokes drag measures the complete surrogate transport discrepancy.
2. PINN flow with ForceNet versus the same PINN flow with exact Stokes drag isolates the force-surrogate contribution.
3. Fluid markers in PINN and FEM fields examine flow-path differences without particle inertia.

The archive contains 119 integrations: 105 particle trajectories and 14 fluid-marker trajectories. They share models and inputs, and are not treated as independent experimental observations.

4.4 Data Provenance and Independence

All current flow, force, and trajectory data are computational. The principal evaluation seed is fixed rather than selected by the smallest FEM error. The two-seed statistics describe observed training variability, with no claim of a confidence interval or population-level inference. The current study is developmental: seeing its failures informs the proposed research plan. A new frozen evaluation set will therefore be used before making final claims about improvements, as specified in chapter 7.

Chapter 5

Verification and Evaluation Design

5.1 Reference Hierarchy

The evaluation proceeds from exact checks to independently discretized comparisons. This hierarchy separates equation implementation, numerical approximation, and physical applicability [8].

For plane Poiseuille flow in a straight channel of half-width h,

$$u = 1 - (y/h)^2, \qquad v = 0, \qquad p = -2x/h^2 + C. \tag{5.1}$$

This profile checks both FEM and a separately trained straight-channel PINN. Velocity profiles are compared on a transverse section; relative pressure errors use points spanning the channel length, because exact pressure has no transverse variation on a single section.

In a uniform flow, the Stokes relaxation solution provides an independent particle-ODE check,

$$\boldsymbol{v}_p(t) = \boldsymbol{u} + [\boldsymbol{v}_p(0) - \boldsymbol{u}] \exp(-t/\mathrm{St}). \tag{5.2}$$

Algebraic drag tests quantify whether ForceNet reproduces its specified law; they do not validate that law against real particles near a wall.

5.2 Field and Trajectory Metrics

The reported relative field error is a discrete sample norm,

$$E_u = \frac{\sqrt{\sum_i \|\widehat{\boldsymbol{u}}_i - \boldsymbol{u}_i^{\mathrm{ref}}\|^2}}{\sqrt{\sum_i \|\boldsymbol{u}_i^{\mathrm{ref}}\|^2}}. \tag{5.3}$$

Pressure uses the same construction after each pressure sample set is mean-centered to remove gauge differences. Interior sampling is in channel coordinates; the resulting norm is not an exact area-weighted integral. Maximum velocity error, boundary RMS error, PDE residuals and section flux provide complementary information. FEM's integrated divergence norm and PINN's sampled continuity RMS have different definitions and should not be compared as if they were identical measures.

For a pair of paths, the position RMS is calculated at 150 common-time samples up to $T_c = \min(T_{\mathrm{PINN}}, T_{\mathrm{ref}})$. Its ratio to width W is reported alongside $T_c / \max(T_{\mathrm{PINN}}, T_{\mathrm{ref}})$ and endpoint-class agreement. Restricting the time interval can hide disagreement after one path ends; the coverage and class measures prevent that truncation from being interpreted as full-path agreement.

Table 5.1: Numerical screening criteria used to organize the proposal.

Quantity	Screening requirement
Channel velocity and pressure	Relative sampled error below 5% each
Straight-channel PINN	Velocity and pressure error below 5% each
Boundary velocity	Dimensionless RMS below 0.02
Section mass flux	Maximum relative deviation below 5%
Reference mesh difference	Penultimate-to-finest velocity difference below 2%
Reference trajectory interpolator	Velocity discrepancy below 2%
Force surrogate	Held-out normalized relative error below 5%
Baseline trajectory position	RMS below 5% of width for every compared seed
Baseline endpoint classes	Agreement for every compared seed
Common trajectory time	More than 95% for every compared seed
Separate ABC benchmark	Velocity and pressure errors below 5% each

These are numerical screening choices, not universal engineering tolerances. The current records are a developmental baseline; this proposal does not claim that the listed thresholds were preregistered before all preliminary inspection. Final acceptance criteria and held-out cases will be frozen before the proposed comparison experiments begin.

5.3 Separate Three-Dimensional Benchmark

The exact ABC field, with unit amplitudes, is

$$\begin{aligned} \boldsymbol{u} &= (\sin z + \cos y,\ \sin x + \cos z,\ \sin y + \cos x), \\ p &= -\mathrm{Re}\|\boldsymbol{u}\|^2/2, \qquad \boldsymbol{f} = \boldsymbol{u}, \end{aligned} \tag{5.4}$$

on $[0, 2\pi]^3$. It satisfies the forced steady momentum equation $\mathrm{Re}(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \nabla^2 \boldsymbol{u} + \boldsymbol{f}$. Indeed, $\nabla \cdot \boldsymbol{u} = 0$, $\nabla \times \boldsymbol{u} = \boldsymbol{u}$, and $\nabla^2 \boldsymbol{u} = -\boldsymbol{u}$. The convective term is the gradient of $\|\boldsymbol{u}\|^2/2$, which is canceled by the pressure gradient.

A separate 3D network predicts all three velocity components and pressure; known velocity is prescribed on the cube faces, and one pressure value fixes the gauge. The exact interior field is reserved for evaluation. Particle-free tracer paths use the periodically extended exact or learned velocity. This tests genuine 3D derivatives and trajectories, but not a finite-depth curved channel with sidewalls. The extruded planar channel illustration has an entirely different role and is identified as visualization only.

5.4 Uncertainty and Physical Validation

The current error budget includes reference mesh and geometry error, interpolator error, PINN approximation, optimizer variability and ODE error. Two seeds and a three-mesh comparison do not constitute a formal uncertainty analysis or a grid-convergence index. Temporal and event-location refinement are required in the proposed work. At present, no measurement dataset has been supplied; physical validation and uncertain particle properties remain open requirements before a device-level claim is justified.

Chapter 6

Preliminary Results and Interpretation

6.1 Completed Evidence

The following observations are completed repository results, not outcomes promised for the proposed study. The archived evaluation completed 35 stages with 0 stage failures and 0 failed integrations. Numerical execution success is separate from satisfying the accuracy criteria in table 5.1. Table 6.1 summarizes the main measurements.

Table 6.1: Automatically extracted preliminary numerical evidence.

Comparison	Metric	Value
Curved channel	Velocity relative error	1.510%
Curved channel	Pressure relative error	0.863%
Drag network	Held-out normalized force error	0.717%
Poiseuille PINN	Velocity relative error	0.217%
Poiseuille PINN	Pressure relative error	0.180%
Poiseuille FEM	Velocity relative error	8.192×10^{-10}
Finest FEM pair	Velocity difference	0.173%
Reference interpolation	Velocity difference	0.177%
Baseline particle outcomes	Matched endpoint classes	6/7
ABC benchmark	Velocity relative error	2.428%
ABC benchmark	Pressure relative error	13.501%

6.2 Analytical and Reference Checks

Figure 6.1 compares analytical, FEM, and PINN straight-channel velocity profiles. The PINN velocity error is 0.217%, while its longitudinally sampled pressure error

is 0.180%. The FEM velocity error is near floating-point/iteration accuracy for this polynomial solution. This verifies the controlled case, not the accuracy of every curved geometry.

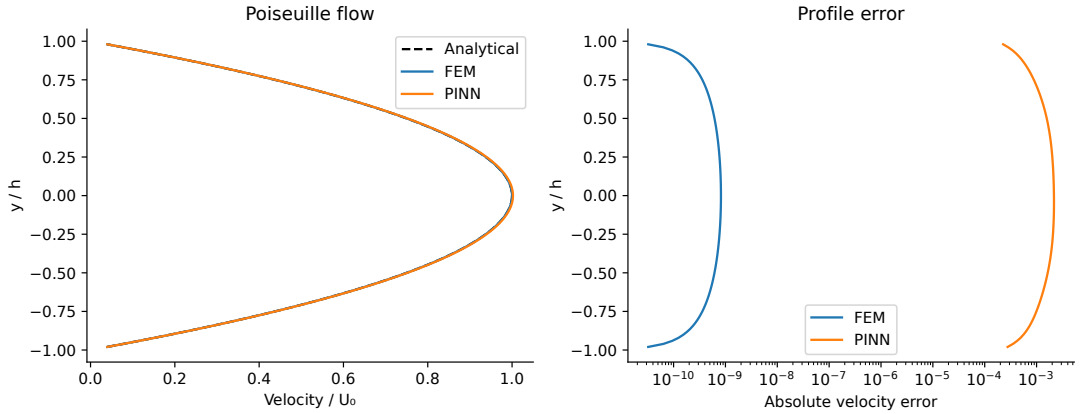


Figure 6.1: Analytical plane Poiseuille profile with independent FEM and separately trained PINN profiles. The second panel shows absolute velocity error. Relative pressure errors are evaluated on longitudinally distributed points, not this section.

Table 6.2: Three reference meshes and their measured sensitivity.

Grid	Triangles	Velocity difference (%)	$\|\nabla \cdot \boldsymbol{u}\|_{L^2}$
30×6	290	1.141	2.448×10^{-1}
60×12	1298	0.173	7.819×10^{-2}
90×18	3026	0.000	4.128×10^{-2}

The penultimate-to-finest velocity difference is 0.173%, and the trajectory interpolator differs from direct FEM velocity evaluation by 0.177%. The zero difference in the finest row is a self-comparison; it is not an estimate of zero discretization error. The declining divergence norm in table 6.2 and the sensitivity plot in figure 6.2 support reference refinement, while leaving systematic modeling error unmeasured.

6.3 Curved-Channel Flow

The principal seed gives sampled velocity and pressure errors of 1.510% and 0.863%, respectively. Figure 6.3 compares the spatial fields and errors; figure 6.4 shows pointwise held-out agreement. Plotting-grid reference values use the measured fast interpolation; numerical validation metrics use direct FEM evaluation. Maximum section-flux deviation is 2.266%, which is below the present 5% screen but remains a nonzero transport inconsistency.

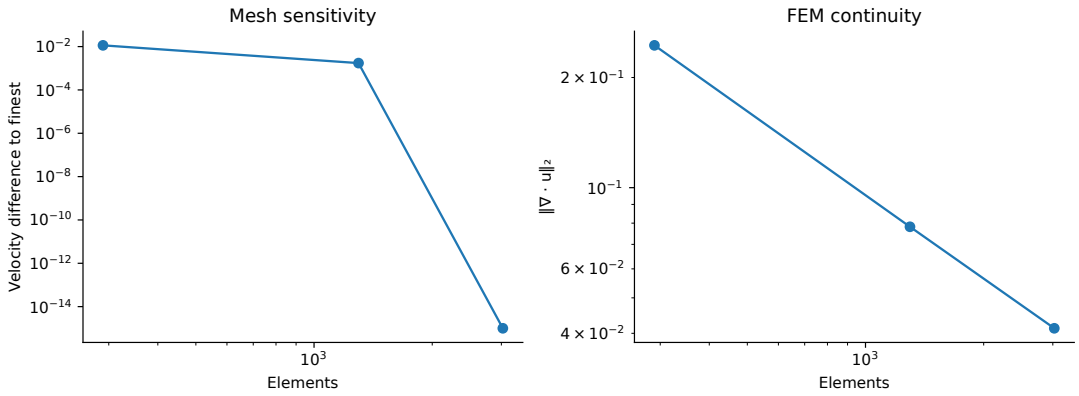


Figure 6.2: Velocity differences to the finest available mesh and the integrated FEM divergence norm. The plotted floor for the finest self-comparison is a display choice, not a formal uncertainty estimate.

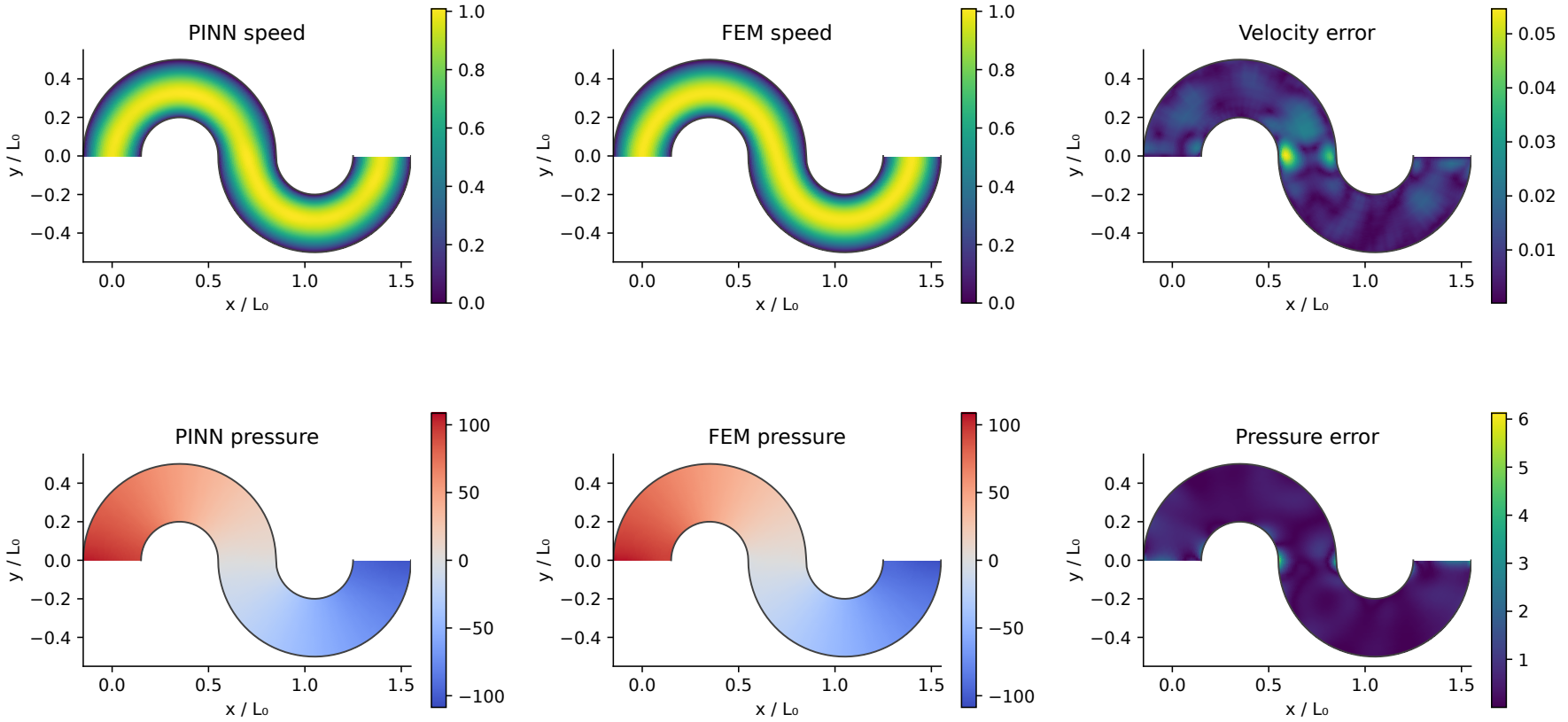


Figure 6.3: PINN and FEM speed and gauge-aligned pressure with local error maps. Both methods solve the same stated boundary-value problem. Smooth appearance does not imply uniformly small boundary or trajectory error.

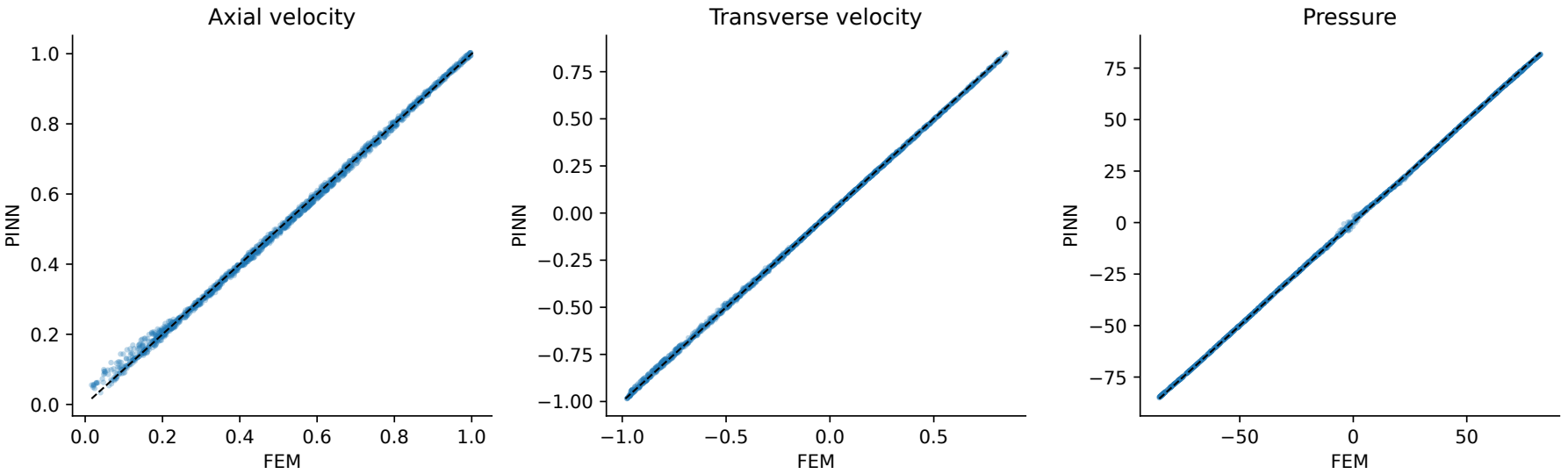


Figure 6.4: Direct FEM/PINN parity on independent interior sample points. Pressure samples are mean-centered before comparison.

Table 6.3: Flow error across the two recorded training seeds.

Seed	Velocity error (%)	Pressure error (%)
42	1.510	0.863
43	1.429	0.886

The velocity error has mean 1.469% and sample standard deviation 0.057 percentage points across these two seeds. The low observed spread does not establish a general training-success probability; the sample is too small for that conclusion.

6.4 Particle Paths and Endpoint Decisions

Figure 6.5 compares the baseline particle paths. Endpoint classes agree for 6 of 7 inlet seeds. The reference predicts exit for all seven; the surrogate predicts sticking for one. In addition, the largest common-time RMS path discrepancy is 34.973% of the channel width. Thus, a field error near 1.5% does not make the baseline particle prediction uniformly accurate.

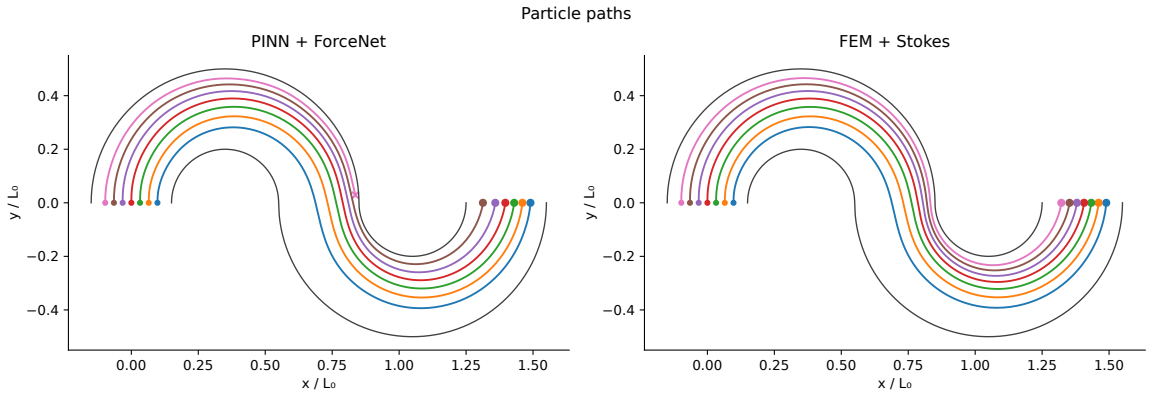


Figure 6.5: Baseline particle trajectories for PINN plus ForceNet and FEM plus exact Stokes drag. Crosses mark sticking; other endpoint types are recorded in the data. The outer-seed endpoint discrepancy is retained.

The lowest coverage is 73.553%, so the matched-time error alone would conceal an appreciable difference in termination time. Table 6.4 motivates retaining all three trajectory criteria. Comparing ForceNet with exact drag on the same PINN flow preserves all seven baseline endpoint classes. That isolation result suggests the flow approximation is a more consequential contributor to the observed class discrepancy in this case, although it does not prove a single local cause.

Table 6.4: Baseline particle comparisons on the common physical time interval.

Inlet η	RMS / width (%)	Time coverage (%)	Same endpoint
-0.650	2.668	98.86	Yes
-0.433	2.299	99.33	Yes
-0.217	4.298	99.27	Yes
0.000	7.834	98.59	Yes
0.217	16.086	96.19	Yes
0.433	34.973	87.39	Yes
0.650	3.295	73.55	No

6.5 Diameter and Density Sensitivity

The path comparisons in figures 6.6 and 6.7 show the effect of the selected inertia and size changes. Table 6.5 gives the actual counts. The largest diameter sticks for every seed in both approaches; the higher-density case retains a disagreement in the number exiting. These observations are conditional on the drag-only model and selected seven starting positions. They are not estimates of a device's population deposition efficiency.

Table 6.5: Endpoint counts in the completed parameter cases; seven seeds per row.

Case	Method	Exit	Stick	Other
Baseline	PINN + ForceNet	6	1	0
Baseline	FEM + Stokes	7	0	0
$d_p = 0.0105$	PINN + ForceNet	7	0	0
$d_p = 0.0105$	FEM + Stokes	7	0	0
$d_p = 0.0255$	PINN + ForceNet	6	1	0
$d_p = 0.0255$	FEM + Stokes	7	0	0
$d_p = 0.045$	PINN + ForceNet	0	7	0
$d_p = 0.045$	FEM + Stokes	0	7	0
$\rho_r = 200$	PINN + ForceNet	7	0	0
$\rho_r = 200$	FEM + Stokes	7	0	0
$\rho_r = 1000$	PINN + ForceNet	6	1	0
$\rho_r = 1000$	FEM + Stokes	7	0	0
$\rho_r = 3000$	PINN + ForceNet	0	7	0
$\rho_r = 3000$	FEM + Stokes	1	6	0

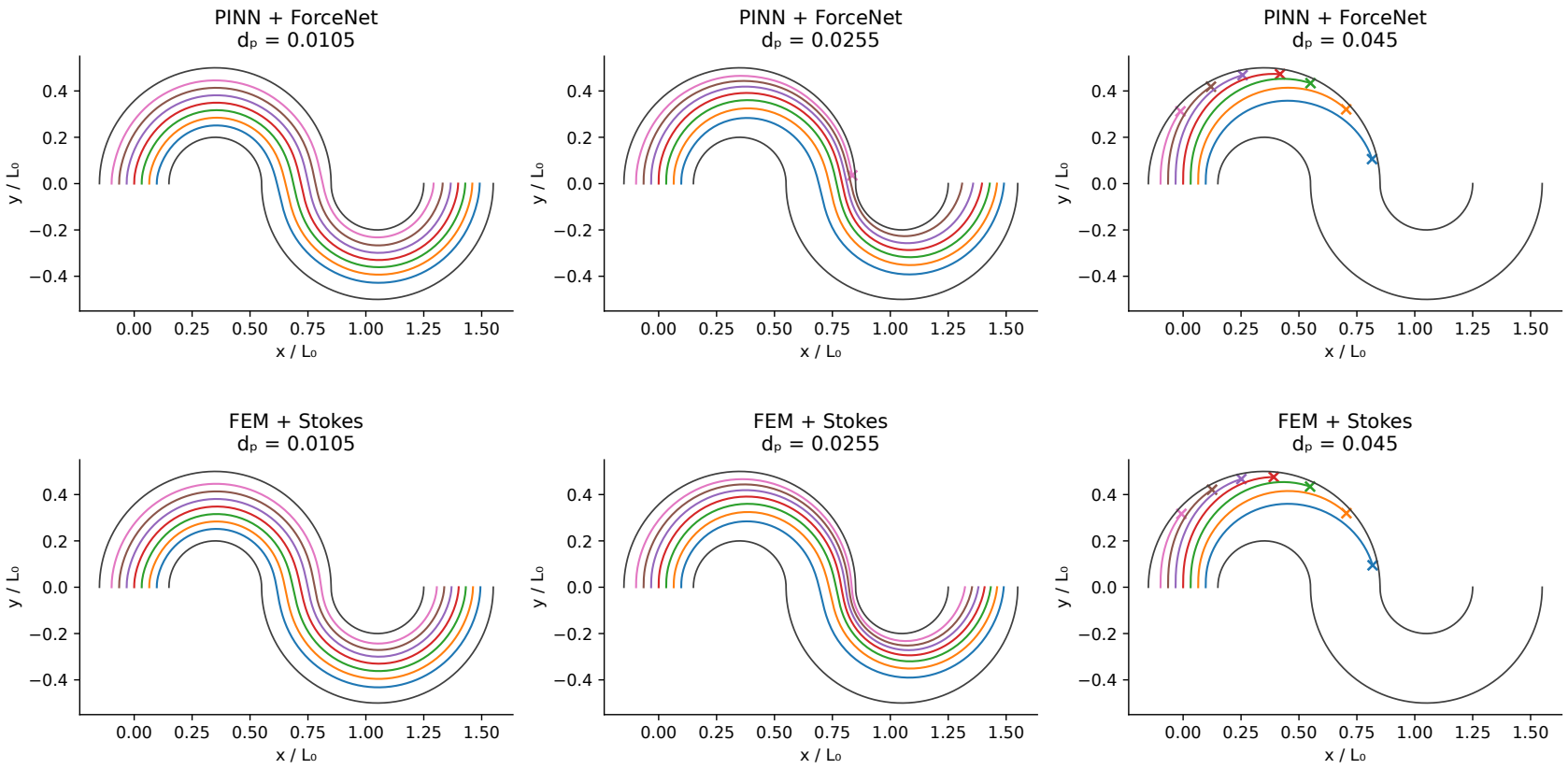


Figure 6.6: Particle paths for the three recorded diameters. The largest diameter is 15% of the width, so physical confinement effects need further consideration.

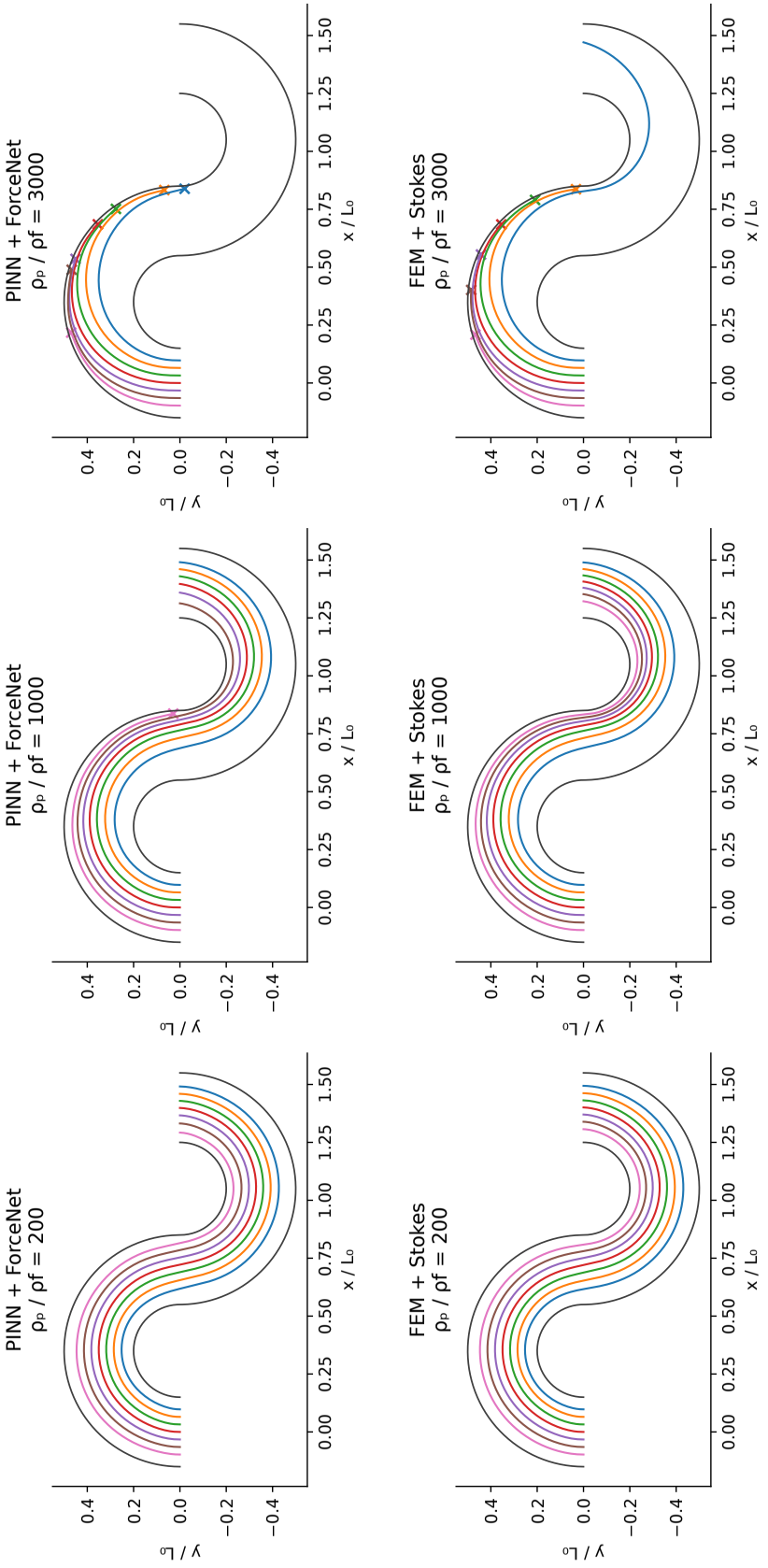


Figure 6.7: Particle paths for the three recorded density ratios under the same stated flow and contact model.

6.6 Three-Dimensional Evidence

The separate ABC model has velocity error 2.428% and pressure error 13.501%. The latter exceeds the current 5% criterion. Figures 6.8 and 6.9 demonstrates nonplanar trajectory integration and records the associated approximation quality; it does not provide a validated 3D curved-channel solution.

The unresolved path accuracy, endpoint agreement and 3D pressure motivate the next experiments in chapter 7. Additional diagnostics, including the planar extrusion labeled as such, appear in chapter B.

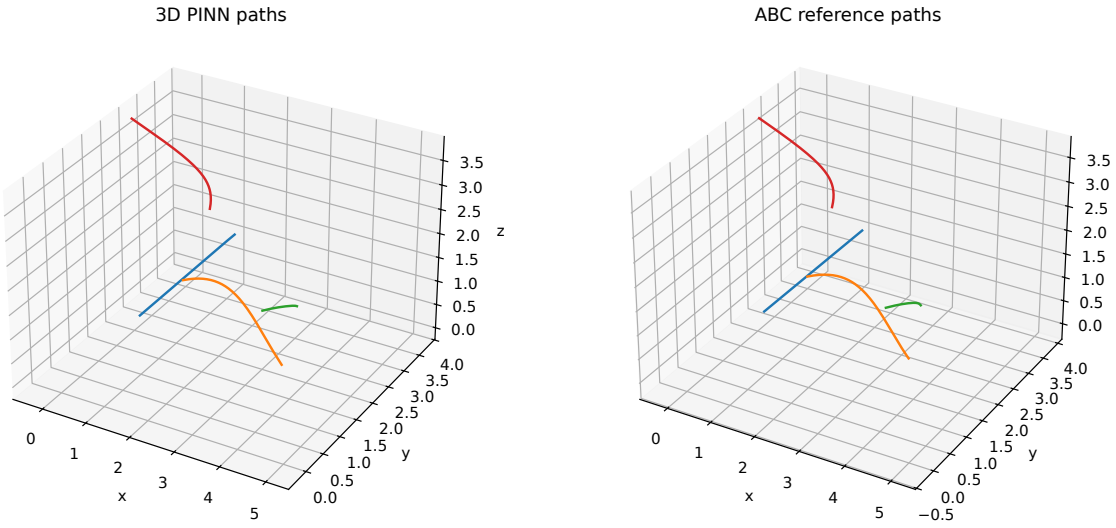


Figure 6.8: Tracer paths from the 3D PINN and exact forced ABC velocity fields. These are genuine three-coordinate paths in a separate benchmark.

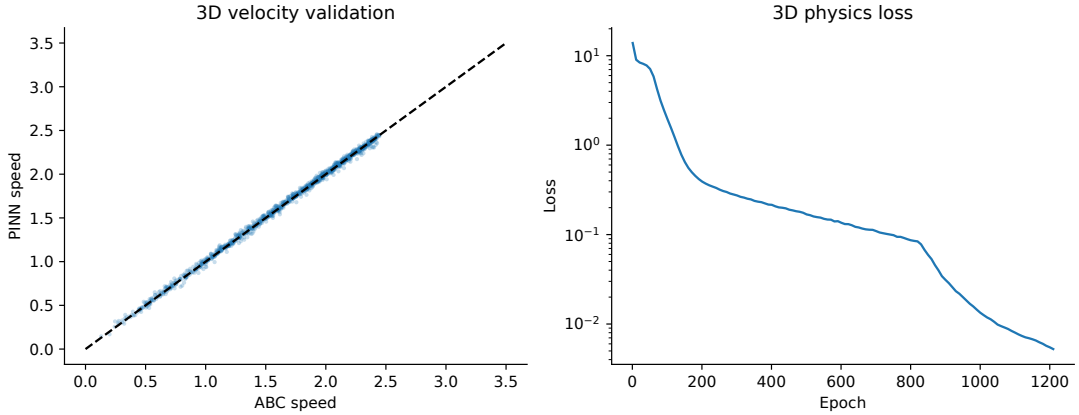


Figure 6.9: Speed parity and physics loss for the ABC benchmark. The pressure error is reported numerically and remains above the current screening criterion.

Chapter 7

Proposed Work, Deliverables, and Feasibility

7.1 Work Packages

7.1.1 Reference and Evaluation Control

First, preserve the present data as the developmental baseline and create new fixed evaluation points and starting-position sets. Check FEM mesh and geometric refinement beyond the present pair where feasible. Repeat selected particle paths with tighter ODE tolerances, smaller maximum steps, and refined wall-event localization. This separates trajectory-integrator sensitivity from flow-surrogate error before retraining is evaluated.

7.1.2 Transport-Directed PINN Improvement

Compare the current physics-and-flux loss with controlled alternatives: increased boundary sampling, targeted collocation near turning regions, and residual-informed sample redistribution. Use equal documented training budgets where possible and record any extra compute. Develop each change on the development set, then freeze the procedure before evaluating held-out locations. The primary endpoints will be path error, time coverage and endpoint agreement, in addition to velocity and pressure norms.

The main force ablation will retain exact Stokes drag on the learned flow. It prevents a flow improvement from being confused with a change in the force surrogate. If a force network provides no measurable benefit for this known algebraic law, exact drag may remain the preferred computational baseline. A future empirical force model would need external force data and a separate identification/validation protocol.

7.1.3 Geometry and Parameter Coverage

Use straight, single-bend, S-shaped and smooth custom-centerline channels at a modest set of Reynolds numbers, initially 0.5, 1 and 3. These are proposed cases, not completed new results. Retrain each case first to establish a reliable per-case baseline. Only a later model trained across multiple parameter cases could support an unseen-parameter test. Held-out centerline shapes and Reynolds numbers must be specified before any parametric-generalization claim is made.

Retain the existing particle-size and density sensitivities, add starting points near observed event transitions, and distinguish low-confinement quantitative cases from larger-particle exploratory cases. Review Re_p, diameter-to-width ratio and loading assumptions in each case. Changes to force physics require new references, not simply more network training.

7.1.4 Three-Dimensional and Physical Extensions

Improve pressure recovery on the ABC verification problem before extending to a finite-depth channel. A 3D channel study will require a mesh with sidewalls, a three-component velocity network, physically specified 3D inlet/outlet conditions and an independently refined 3D CFD reference. The planar extrusion will continue to be identified as a visual aid.

Physical validation is conditional on an accessible, documented measurement or benchmark dataset with compatible geometry, operating conditions, particle properties and contact behavior. If such data remain unavailable, the final claim will be numerical verification and model-to-model agreement, with experimental validation explicitly outstanding.

7.2 Schedule and Milestones

The following relative schedule begins after project approval; it does not imply an institutional deadline or an approved research start date.

7.3 Resources, Risks, and Decision Rules

The current workflow runs on CPU with bounded training and integration budgets. GPU use may reduce training time, but exact speedups and total future compute requirements have not been benchmarked. Separate costs will be recorded for training, reference assembly/solve, trajectory evaluation and reporting; reloading weights is not a new training-time measurement.

Table 7.1: Proposed six-month research schedule.

Period	Main activity	Reviewable milestone
Month 1	Freeze development/evaluation separation; refine references and ODE events	Reference and temporal sensitivity record
Month 2	Boundary and turning-region training ablations	Controlled field and trajectory comparison
Month 3	Expand starting positions, particle cases and force isolation	Transport-error and endpoint-transition maps
Month 4	Separately trained geometry/Reynolds-number study	Case matrix with explicit supported scope
Month 5	Improve ABC pressure; define a bounded 3D channel extension	3D verification assessment and feasibility decision
Month 6	Consolidate uncertainty, reproducibility and available physical evidence	Final analysis, thesis/proposal updates and research manuscript outline

Table 7.2: Principal research risks and responses.

Risk	Response and decision rule
Low global error but wrong particle event	Add local and event-based diagnostics; withhold a transport-accuracy claim until path criteria are met.
Reference or event error dominates	Refine the reference and time/event resolution before attributing the discrepancy to the PINN.
Unstable or costly training	Compare fixed budgets and multiple seeds; retain failed cases and report supported operating ranges.
Particle-model assumptions fail	Restrict quantitative claims or introduce and independently validate additional physics.
Experimental evidence is unavailable	Limit the final validation statement to numerical comparisons.
3D pressure remains inaccurate	Treat the benchmark as unresolved and defer broad 3D channel claims.

7.4 Expected Contributions and Deliverables

The proposed deliverables are a validated numerical reference hierarchy, documented relationships between field and transport error, a controlled geometry/parameter evaluation, and a reproducible release containing configuration, trained weights, raw data, figures and a data-driven LaTeX report. These are research targets whose success will be judged using the frozen evaluation design. The present proposal already demonstrates the reporting mechanism and identifies where the preliminary model needs improvement; it does not assume that every proposed target will succeed.

No human or animal data are used in the present numerical work. Any later experimental collaboration, data reuse, institutional submission or required authorship/assistance disclosure will follow the applicable institutional and dataset terms. The current repository does not establish a separate third-party data license or an experimental approval.

References

[1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019. doi:10.1016/j.jcp.2018.10.045.

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[6] T. Gustafsson and G. D. McBain, "scikit-fem: A Python package for finite element assembly," *Journal of Open Source Software*, vol. 5, no. 52, article 2369, 2020. doi:10.21105/joss.02369.

[7] M. R. Maxey and J. J. Riley, "Equation of motion for a small rigid sphere in a nonuniform flow," *The Physics of Fluids*, vol. 26, pp. 883–889, 1983. doi:10.1063/1.864230.

[8] P. J. Roache, *Verification and Validation in Computational Science and Engineering*. Albuquerque, New Mexico: Hermosa Publishers, 1998.

Appendix A

Reproducibility and Evidence Record

A.1 Source and Software

The numerical inputs are from repository data revision `97d71480fc01`. Figure/data SHA-256 hashes are written to `generated/evidence_manifest.json` when the document is built. The archived numerical run retains its own source hashes, configuration, model checksums and stage records. A later report revision does not change those underlying predictions.

Table A.1: Software versions recorded with the numerical evidence.

Package	Recorded version
numpy	2.3.5
scipy	1.17.0
torch	2.14.1+cpu
matplotlib	3.10.8
scikit-fem	12.0.2
Pillow	12.3.0

The scientific code was exercised under Python 3.12. These are numerical-run versions, not a claim that a PDF build retrained the models. The proposal builder itself uses the Python standard library and a standard LaTeX installation; PyTorch is not needed simply to compile this document.

A.2 Building the Proposal

From a checkout of the repository, run:

```
python proposal/build_proposal.py --bundle
```

The builder regenerates tables and highlighted values, checks required inputs, runs latexmk/pdfLaTeX to resolve references, and writes `proposal/Particle_PINN_Proposal.pdf`. The optional portable source ZIP includes the numerical inputs and all required figure PDFs. The original user-supplied template is preserved under `proposal/template/`. For a new numerical run, use:

```
python proposal/build_proposal.py --results results/NEW_RUN
```

The schema and required benchmarks must match; missing data trigger an explicit error rather than numerical substitutions. Generated tables follow the selected data. The interpretive prose is a reviewed narrative and must also be checked when the scientific case or findings change.

A.3 Inspecting and Reproducing the Numerical Study

The following commands inspect the existing tests and re-evaluate saved weights using the repository workflow. Install the scientific dependencies as described in the project README before running them.

```
python -m unittest discover -s tests -v
python run_particle_pinn.py --preset standard --seeds 2 \
  --ns 60 --nn 12 --weights-from results/final_study \
  --out results/proposal_replay
python assess_results.py results/proposal_replay
```

The replay writes a fresh output directory. It evaluates existing weights; it does not reproduce the original training time. Model provenance paths inside the old metadata are historical records; the required weights are present under `results/final_study/models/`. New training is performed with the documented training options rather than the replay command.

For an interior fluid marker with an arbitrary admissible start,

```
python trace_path.py --run results/final_study \
  --start 0.35 0.35 --tracer --out results/interior_tracer
```

The equivalent particle command omits `-tracer`; an initial velocity can be supplied explicitly. A new geometry or Reynolds number requires retraining under the present per-case design.

A.4 Complete Numerical Screening Record

Table A.2 is generated from the stored criterion flags. These flags retain distinctions between execution, field accuracy and transport accuracy. They are numerical screens, not an institutional approval or experimental validation statement.

Table A.2: Recorded numerical screening outcomes.

Check	Outcome
Analytical FEM	Met
Poiseuille PINN velocity	Met
Poiseuille PINN pressure	Met
FEM nonlinear solve	Met
Reference mesh sensitivity	Met
Reference interpolator	Met
Channel velocity	Met
Channel pressure	Met
Velocity boundary	Met
Channel mass flux	Met
Drag surrogate	Met
Trajectory integration	Met
Particle endpoint agreement	Not met
Particle position error	Not met
Trajectory time coverage	Not met
ABC velocity	Met
ABC pressure	Not met
Execution stages	Met

A.5 Availability and Review

The repository contains code, weights, raw numerical outputs, vector figures, PNG images, a GIF and result reports. The animation is available as the particle-motion GIF. A PDF figure cannot encode that animation, so the underlying paths are also shown as static figures and retained in CSV form. Access follows the repository's current GitHub permissions. The supplied thesis template is retained as a user-provided formatting source; no broader reuse license or institutional approval is inferred from it.

Appendix B

Supplementary Numerical Figures

All panels in this appendix are existing repository exports. They supplement the principal evidence without changing the physical interpretation. Loss and seed plots support optimization diagnosis; flux and profiles test the flow; force and trajectory plots help separate modeling components. Figure B.5 is explicitly a planar extrusion, not the separate 3D benchmark in figure 6.8.

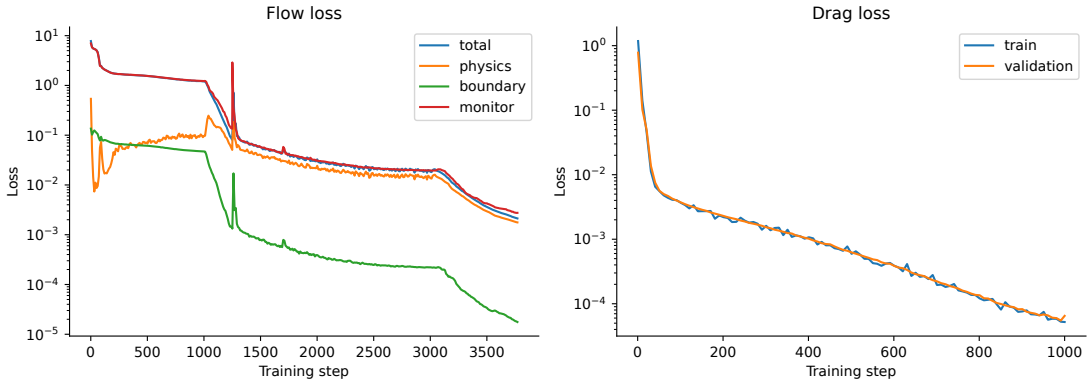


Figure B.1: Flow and force loss histories. Training-step labels include the recorded optimizer phases; a low objective alone is not a transport-accuracy certificate.

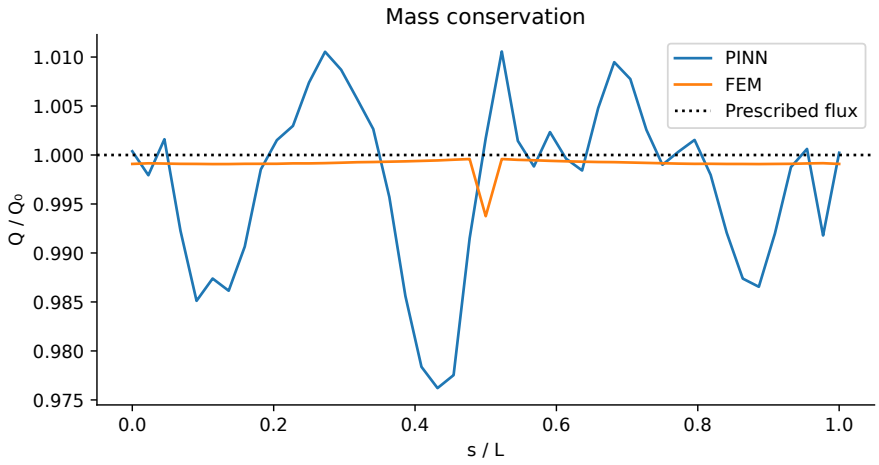


Figure B.2: Computed section flux normalized by prescribed flux. The departure from unity is measured separately from field error.

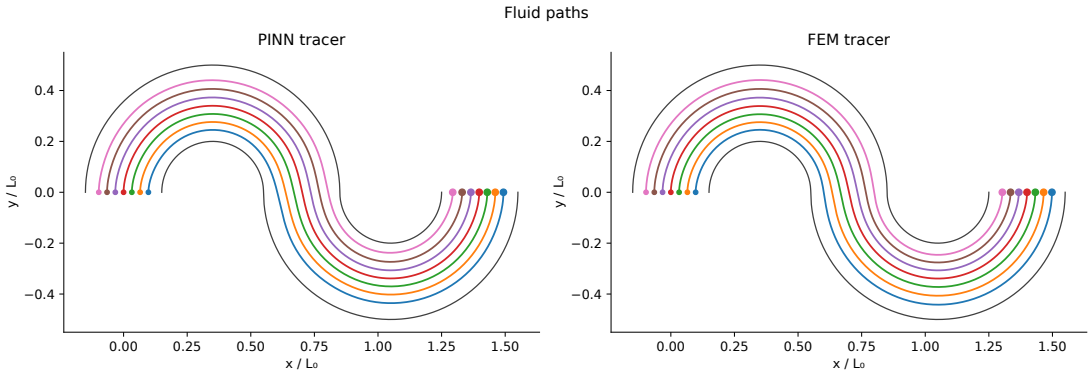


Figure B.3: Fluid markers integrated in the separately labeled learned and reference velocity fields.

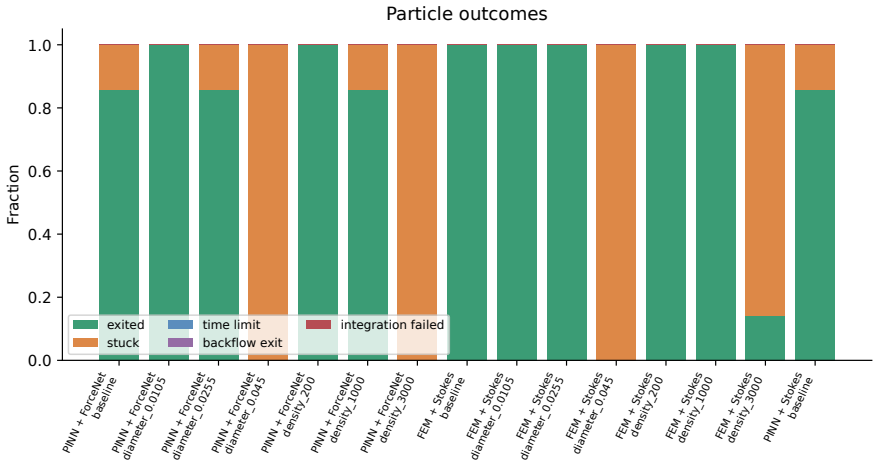


Figure B.4: Endpoint-class fractions for each recorded case. Numerical failures, if present, would remain a separate class. Exact case counts are in table 6.5.

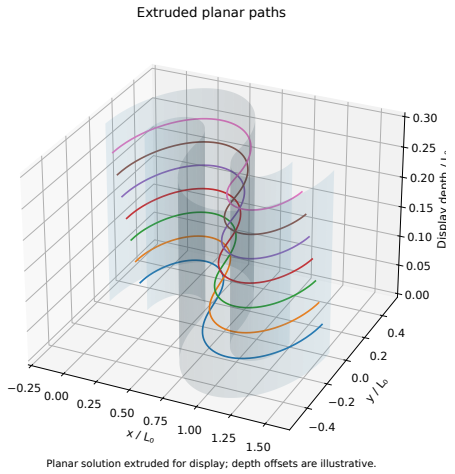


Figure B.5: Extruded visualization of planar particle paths. The displayed depth offsets are illustrative and do not come from a three-dimensional channel solution.

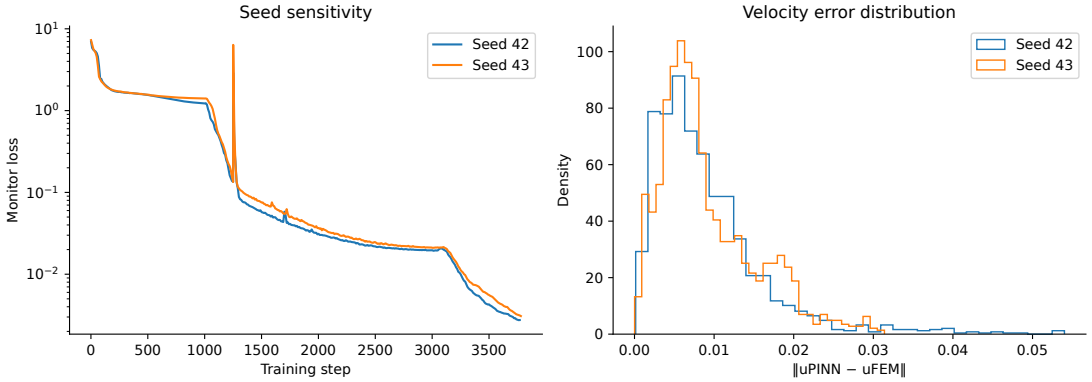


Figure B.6: Monitor-loss histories and held-out velocity-error distributions for the two recorded seeds.

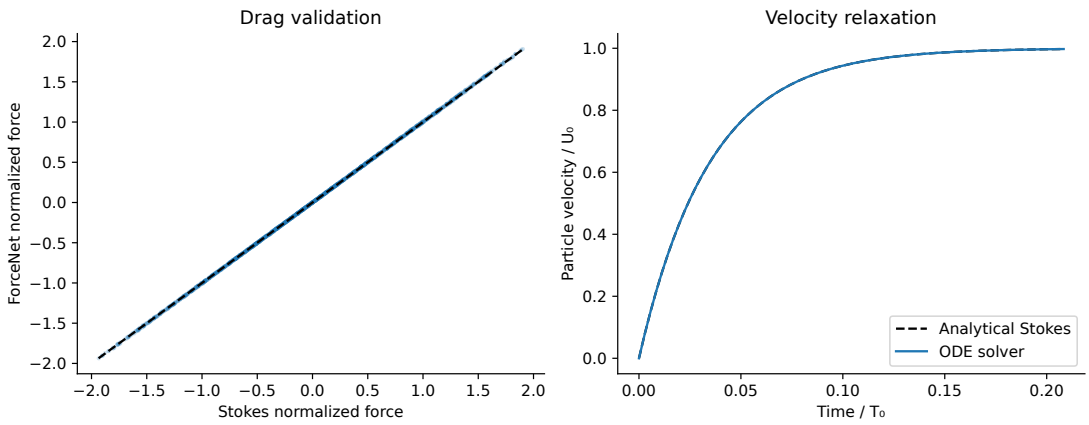


Figure B.7: Normalized ForceNet/Stokes parity and a numerical ODE comparison with exact Stokes velocity relaxation.

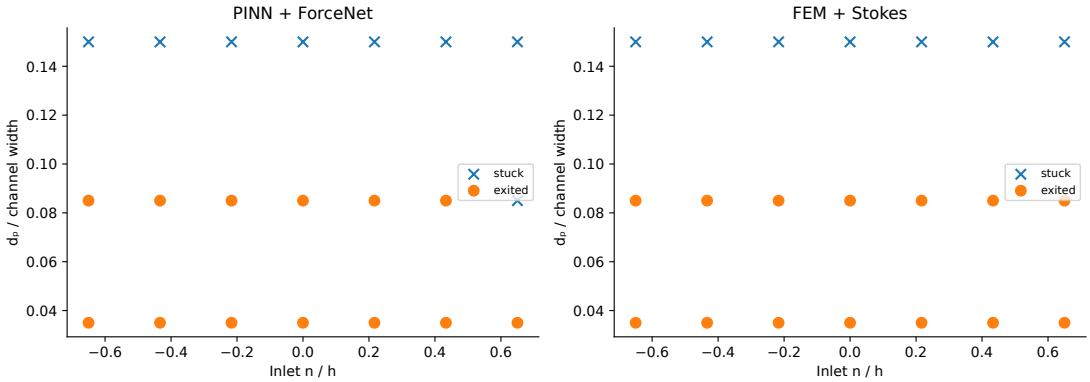


Figure B.8: Endpoint classes versus inlet position and diameter-to-width ratio. The map uses the stated sticking model and the available finite set of seeds.

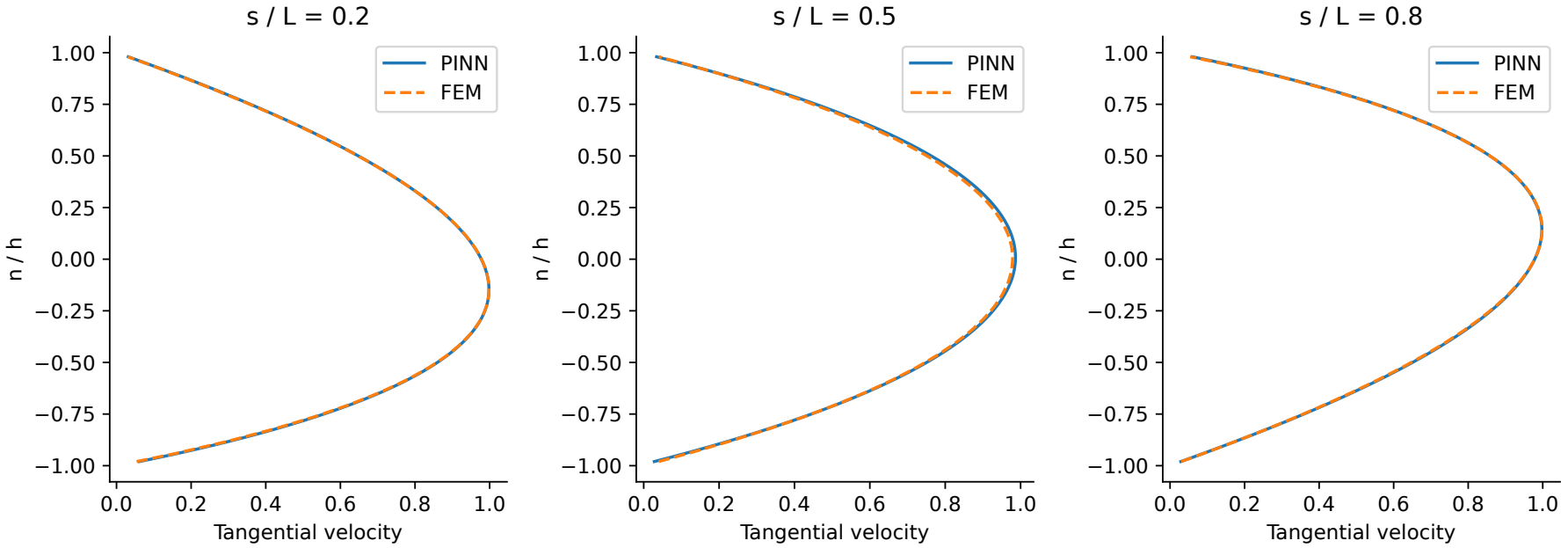


Figure B.9: Tangential velocity profiles at three channel locations, comparing PINN and the reference field.

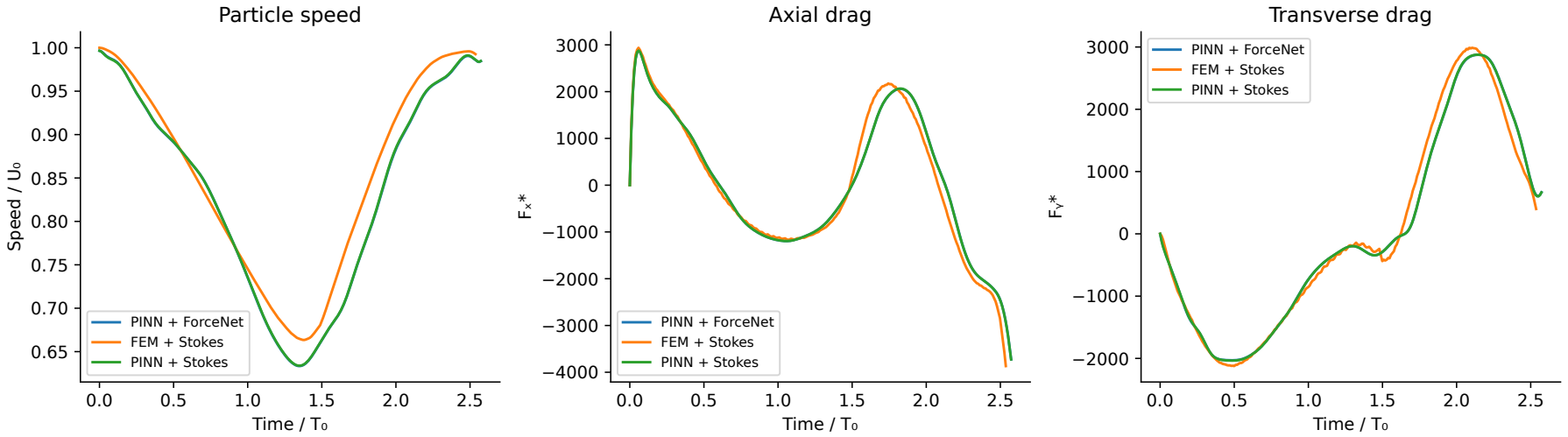


Figure B.10: Central-seed baseline particle speed and drag components. The displayed forces are hydrodynamic drag; wall-contact constraint forces are outside this model.